2

CHAPTER

Natural Resources and Associated Problems

Chapter Outline

- *Introduction* • *Types of Natural Resources* • *Depletion of Natural Resources*
- *Conservation of Natural Resources* • *Summary* • *Study Questions*

INTRODUCTION

The natural resource is the stock that can be drawn from nature. The earth is a storehouse of various resources, such as land, water, soil, minerals, vegetation, forests, fuels, solar energy *etc.* that are of immense importance to mankind. Thus, natural resources are the components of the environment that can be drawn upon for supporting life. Any such material which can be used directly or after transformation to sustain life is resource. Without these natural resources, life on earth would not have been possible. Man satisfies his primary needs of food, shelter and clothing and fulfils his other requirements with the resources made available to him from nature.

The term '**resource**' is dynamic but its meaning got changed with the advancement in science and technology. Previously, natural resources were defined as those materials which were of value to a particular human culture (Dasmann, 1968). But, now everything on this earth is useful to man. Some of the resources, such as land, water, air, light, plants and animals are readily available, whereas others, such as minerals, coal, oil *etc.* are hidden under earth's surface and are to be taken out.

TYPES OF NATURAL RESOURCES

The earth's natural resources can be categorised on the basis of their chemical nature, availability or abundance, area of occurrence, origin and utility *etc.*

I. Natural Resources Based on Chemical Nature

Based on chemical nature, natural resources may be of the following types:

1. **Inorganic Resources:** These natural resources are inorganic in nature. Inorganic resources include air, water, rocks, mineral ores, minerals *etc.*
2. **Organic Resources:** These natural resources are organic in nature. Organic resources include organisms, such as plants (forests), animals, microbes and their products like fossil fuel (oil, coal, natural gas *etc.*).
3. **Mixed Resources:** These natural resources are a combination of both inorganic and organic components. Soil is the best example of mixed resource as it is formed of both inorganic components, *i.e.,* inorganic matter of the soil which results from weathering of rocks and organic components, *i.e.,* humus from the plants and animals formed by the activity of microorganisms.

II. Natural Resources Based on Availability or Abundance

Based on availability or abundance natural resources may be of the following types:

1. **Inexhaustible Resources:** These natural resources are found in such abundance so as to unlikely deplete or get exhausted. They are not likely to be exhausted by man's consumption. Inexhaustible resources include solar energy, air, water, sand, clay, tidal energy, hydropower, wind, temperature, precipitation *etc.*
2. **Exhaustible Resources:** These natural resources are likely to be depleted and then exhausted upon their continuous exploitation as they have a limited stock on the earth. Exhaustible resources may be non-renewable or renewable.

   (a) Non-renewable Resources: Non-renewable natural resources lack ability of recycling and replacement or are replaced after a very long time. They are unevenly distributed in the world. Some are found abundant in one country and poorly present or even absent in another country. Similarly a country may be rich in one kind of resource and poor in other kinds of resources. Non-renewable resources may be natural, *e.g.* fossil fuels or man made, *e.g.* polythene. Naturally occurring non-renewable resources include mineral ores, minerals, metals and fossil fuels. These occur in earth's crust and are difficult to be taken out. Fossil fuels, such as coal, oil, natural gas have a limited stock and require millions of years to reform. Their consumption rate is very high. At this rate of consumption

it is likely that these may be exhausted before they are reformed. They cannot come back to their entity after being used and cause severe pollution. Minerals are also being used at a very fast rate. The reformation period of minerals and fossil fuels is very long. Similarly, soil is often considered as non-renewable resource as it can be degenerated only in few years but need several hundred years for its formation. Genetic biodiversity is also a non-renewable natural resource. Care should be taken in use of these resources so that they may not be exhausted and that nothing is left for future.

(*b*) ***Renewable Resources:*** Renewable natural resources can be replenished, reproduced or recycled. They have the inherent capacity of reappearing or replacement within a reasonable time and maintain themselves. They can last indefinitely. A judicious balance is required to be maintained between their exploitation and replenishment. Over-use or improper management can diminish or exhaust renewable resources. The chief renewable resources are soil fertility, water and living organisms: plants (forests), animals and microorganisms. If consumption of these resources exceeds their rate of renewal, not only their quality becomes affected but they may become non-renewable. For example, excessive felling of trees results in deforestation and overgrazing causes denudation of the land and desertification. Over consumption of ground water may lead to lowering of water table. Thus, these resources need judicious management and conservation so that future generations may not be deprived of them.

III. Natural Resources Based on their Occurrence

Based on occurrence, natural resources may be of the following types:

1. **International Resources:** These natural resources are available to all countries and have no boundaries. Atmosphere, with sunlight and air including many gases, is common to all countries.
2. **Multinational Resources:** These natural resources are shared by more than one country. Certain rivers and lakes may be shared by more than one country, *e.g.* Brahmaputra river is common to Tibbet, India and Bangladesh. The migratory birds also form multinational resource.
3. **National Resources:** These natural resources are restricted to a particular country. Physical features, geological structure, river systems, flora, fauna and human resource constitute the national resource of a country.

IV. Natural Resources Based on their Origin

Based on origin, natural resources may be of the following types:

1. **Biotic Resources:** Biotic resources are basically organic in nature. These include forests and forest produce, vegetation with variety of plants, wide variety of animals, such as birds, fish and other marine life forms, and microorganisms. Fossil fuels also belong to this category since they originate from organic matter. Some biotic resources, such as plants (forests) and livestock are renewable, whereas fossil fuels are non-renewable natural resources.
2. **Abiotic Resources:** Abiotic resources are basically inorganic in nature. They are composed of non-living matter. Land, water and minerals, such as iron, copper, lead, gold *etc.* are common examples of abiotic resources. Some abiotic resources occur as nodules, such as nodules of copper and manganese.

V. Natural Resources Based on their Utility

Based on utility, natural resources may be forest resource, water resource, food resource, energy resource, land resource or any other resource which may be useful for man.

Certain resources, such as energy resources belong to several categories and can be inexhaustible, *e.g.* solar energy, exhaustible renewable, *e.g.* forests and exhaustible non-renewable, *e.g.* fossil fuels-coal, oil, natural gas *etc.*

3

CHAPTER

Forest Resources

Chapter Outline

- *Introduction* • *Forest Distribution* • *Importance of Forests* • *Deforestation*
- *Conservation of Forests* • *Indian Forest Scenario* • *Summary* • *Study Questions*

INTRODUCTION

Forests are one of the most abundant natural resources on this earth. They not only produce innumerable material goods, but also provide various environmental services that are essential for life. The epic 'Vedas' and 'Upanishads' were composed of sages singing *may the god, the water, the plants and the forests, the trees accept our prayers.... May the blessings of our 'simul' (tree) protect us for ever*. The 'Puranas' have said *one tree is equal to ten sons because tree means water, water means bread and bread is life*. According to 'Chandogya' Upanishad, a book thousands of years old *water is the essence of earth and plants are the essence of water*. Gunnar Poulson quotes, *These are useful to man in two different ways: as producers of a wide variety of goods, commonly called 'forest produce', and as custodians of favourable environmental conditions. It would not make sense to try to qualify one of these functions as more important than the other. Both are indisputable to all well beings, indeed to the survival of man*. UNO had declared the year 2011 as International Year of Forests.

Perhaps, the word 'forest' has been derived from the Latin word *Floris*, which means outside, *i.e.*, uncultivated area covered with tall and dense tree growth outside the settlement. In general, the term **forest** is referred to an area, set aside for production of timber and other produce. Ecologically speaking, the forest is a biotic community predominantly of trees and other vegetation. United Nations Food and Agriculture Organisation (FAO) has defined forest land as *all lands bearing vegetation dominated by trees of any size, exploited or unexploited, capable of producing wood or other forest products.*

FOREST DISTRIBUTION

Forests contain about 90 per cent of biomass and cover about one-third of the land area of the earth. Nearly 50 per cent of total forest land is tropical forest. The FAO estimates world's land area as on 1994 to be 144.8 million sq km (or about 29 per cent of the surface of the globe), of which forests and woodlands account for 30 per cent (Fig. 3.1). The FAO defines forest and woodland as land under natural or planted stands of trees, whether productive or not, including land from which forests have been cleared but will be restored in near future. Thousands of years ago, before large-scale human disturbances of the world began, forests and woodlands probably covered about 6.0 billion hectares of land. Since then, about 16 per cent of forest land area has been converted into cropland, pasture, settlements or unproductive wastelands. The FAO estimates the world's area under forests and woodlands as on 1994 to be about 4.7 billion hectares. Closed forests occupy about $4/5^{th}$ of the forest and the rest are open forests. Closed forests are those in which tree crowns spread over 20 per cent or more of the ground and have potential for commercial timber harvesting. Open forests or woodlands are those where tree crown covers less than 20 per cent of the ground.

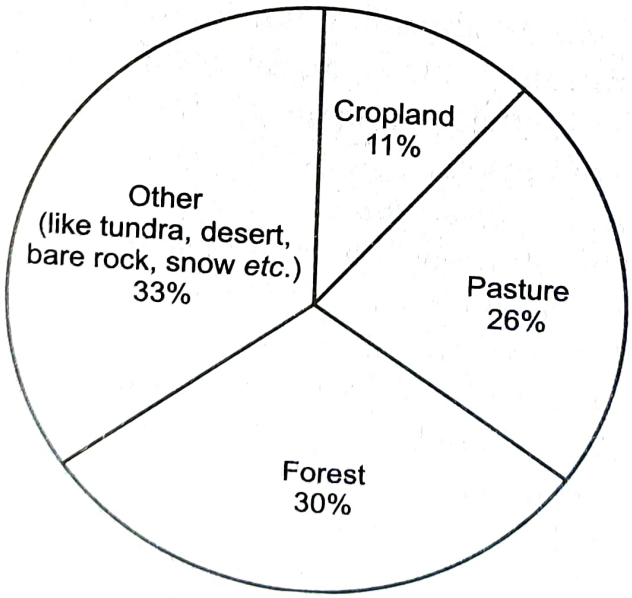


Fig. 3.1. World land use distribution.

World distribution of forests by regions depicts (Fig. 3.2) that Latin America and Russia share between themselves about 39 per cent of world's forests. Africa possesses about 33 per cent forest land. North America and Europe share 11 per cent and 3 per cent, respectively. Asia with almost half of the world's population shares only 14 per cent and India with about

16 per cent of world's population possesses only 1.5 per cent of forest land. Over the years, there has been a decline in forests world over. The greatest loss has been in tropical Asia where 1/3rd of the forests have been destroyed. India lost some 26,000 sq km of dense forests in between 2001-2003.

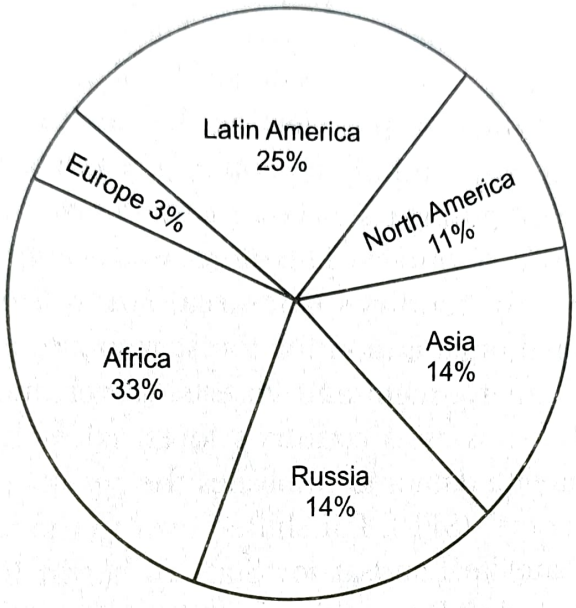


Fig. 3.2. World forests by region.

Distribution of Forests in India

India had abundant forests in ancient periods of Chandra Gupta Maurya who took keen interest in the management of the forests. Ashoka's stone edicts prescribed tree planting. At the beginning of twentieth century, about 30 per cent of land in India was occupied by the forests. But by the end of twentieth century the forest cover declined to 19.4 per cent. This is much less than the goal of having 33 per cent forest cover (20 per cent for the plains and 60 per cent for the hills) as inundated in the National Forest Policy (1988). Dense forest cover, with tree canopy covering 40 per cent or more of land area, decreased tremendously and by 1997 accounted for only 11.5 per cent. It has gone down further and is posing an alarming situation. Of the existing forests, about 65 per cent are dense forests and rest are open degraded forests. Presently, per capita forest area available in India is 0.06 hectares, which is much below the world's average of 0.64 hectares per person. Among sixteen different types of forests in India, composition wise, tropical dry deciduous forests occupy 41.87 per cent, tropical moist deciduous 19.73 per cent and rest are of other types of forests. Open forests and mangrove forests occupy 7.8 per cent and 0.1 per cent, respectively. Nearly 96 per cent of the forests are owned by the government, 2.6 per cent by corporate bodies and the rest are in private sector.

Components of Forest

A forest is essentially a community—an ecosystem with social grouping of plant species in which trees are dominant part of vegetation but other organisms and the soil, water and atmosphere play an important role. Thus flora (vegetation) and fauna (animals)—the biotic components and soil, water and atmosphere—the abiotic components are integral parts of a forest. The biotic components, *i.e.*, flora including variety of plant species and fauna including animals of all kinds, such as mammals, birds, reptiles, insects *etc.* interact with each other and in turn with abiotic components, *i.e.*, air, water, soil, temperature, humus, rain, humidity, wind *etc.* and maintain a balance.

IMPORTANCE OF FORESTS

Forests play a significant role in maintaining the balance in nature. Gautam Buddha quotes *A tree is a peculiar organism of unlimited benevolence. It makes no demand for its sustenance and extends generously its products of life activity. It affords protection to all beings, offering shade even to the axeman who destroys it.* Forests provide important environmental services to mankind and are storehouse of biodiversity. They are the home of innumerable species of plants and animals. Forests have also housed some of the oldest indigenous tribals and played a vital role in the life of forest dwellers. Their main role, in the nature, is to maintain gaseous balance. The services rendered by forests fall under three categories.

I. Productive functions

Productive functions of forests include production of wood and variety of produce which are important for sustenance of life. Forests are of immense importance to the life and prosperity of human beings and the nation.

1. Production of Fuel Wood: Fuel wood is the major forest produce useful to many affluent societies and the rural poor. In developing countries forests are looked upon mainly for the fuel wood. In fact, fuel wood accounts for almost half of all wood harvested worldwide. About 1.5 billion people largely depend on fuel wood for their primary energy requirement. About 58 per cent of total energy used in Africa and about 42 per cent in South-East Asia comes from fuel wood. The world consumption of fuel wood is estimated to be 2,600 million cubic meters till 2005. In India, fuel wood requirement in the year 2000 was 320 Mm^2 and 350 Mm^2 upto 2010.

2. Production of Industrial Wood: The unprocessed logs, the round wood and timber obtained from forests is used for commercial purposes, such as construction of houses, furniture, ship building, boat making and in making of carts, carriages, railways poles, agricultural implements,

packing cases, sports goods, toys *etc*. The wood also finds its use in making paper, plywood, lumber, veneer and other articles of industrial importance. Together, all these purposes account for the figure approaching one-half of the worldwide wood consumption. The developed countries produce less than half of industrial wood but account for about 80 per cent of its consumption. On the other hand, less developed and underdeveloped countries, mainly tropical countries, produce more than half of industrial wood but use only about 20 per cent of it. Important forest wood based industries are plywood, furniture, paper, match box, sports goods *etc*. The FAO estimates that total removal of industrial wood from global forests in 1991 was 3,429 million cubic meters.

3. **Production of Non-wood Produce:** Forests also contribute in production of many non-wood produce which play a significant role in man's life and economy of the country. Vast varieties of food article, fibre, honey, medicinal plants, spices, resins, tannins, gums, lac and many articles of minor importance such as *kutch, ratti, ritha, sikakai, tendu patta, rudraksh* are produced by the forests. Bamboo, commonly called as poor man's wood, is also obtained from the forests. Some forests are also source of minerals. Non-wood produce fetches a big business in the country.

4. **Other Contributions:** Forests provide a wide variety of employment to local people by way of engaging themselves in many forest oriented work plans and industries. Forests add to our knowledge by way of education, research and scientific study. Many of them protect our cultural heritage. Besides, the forests give us aesthetic pleasure and spiritual solace.

II. Protective Functions

Protective functions of the forests include conservation of soil and water; prevention from flood, drought and pollution; and protection against wind, cold, radiation *etc*.

1. **Control of Floods, Drought and Soil Erosion:** Forests control some of the recurrent natural disasters, such as floods, landslides and avalanches. The roots of these plants keep the soil intact. The aerial parts of the plants intercept rains, decreasing their erosive power. Ground flora and thick layers of litter and humus in the forests act as sponge and retain the water received in the form of rains. By decreasing the velocity of water coming down from upper streams, forests help in absorption of water by soil, preventing floods. Extensive roots of the forest trees not only keep the soil particles intact but also induce structural development in the soil by adding humus. After being intercepted by aerial parts of the plants, the rain water reaches the ground level where

it is easily absorbed by the soil. Thus trees along rivers, streams and lakes prevent bank erosion and silt deposition. Importance of forests also lies in preventing droughts leading to desertification.

2. Prevention from Pollution: Forests absorb many toxic gases and help in keeping the air pure and fresh. They operate carbon cycle and convert carbon dioxide into oxygen which is needed for all living beings in the forests and elsewhere. Many trees act as dust filters. Forests are the best tool for abatement of intolerable and high level of noise pollution which has become synonymous with the present day civilised societies with increasing number of automobiles on roads and industries all round. Properly placed screens of trees and shrubs decrease traffic noise along busy streets and highways. Many trees prevent and reduce water pollution in rivers, streams, lakes, reservoirs and estuaries.

III. Regulatory Functions

Regulatory functions of forests include absorption, consumption and release of gases like oxygen and carbon dioxide and minerals, water, radiant energy *etc.* Forests bring about gaseous exchange by releasing oxygen and consuming carbon dioxide. They also bring about moderation of climate and improve temperature conditions and enhance the economic value to landscape. Forests also provide habitat to tribals and wildlife.

1. Maintenance of Gaseous Balance: The main role of forests in nature, is to maintain gaseous balance. All breathing organisms, including man, take oxygen from the atmosphere and release carbon dioxide. Increasing population coupled with increasing number of automobiles, trains, aeroplanes and industries and many other activities release carbon dioxide and other pollutants in the atmosphere. Forests act as 'lungs' as they convert carbon dioxide into oxygen through photosynthesis. A mature *Ficus religiosa* (Peepal) tree, is capable of absorbing 2252 kg of carbon dioxide and releasing 1772 kg of oxygen per hour. Therefore, to save us from carbon dioxide and for maintenance of healthy gaseous balance in the atmosphere still more green plants (forests) are needed.

2. Habitats of Flora and Fauna Forests are important habitats for a large diversity of flora and fauna. They have also housed and are still housing some of the tribals. They play a vital role in the life and the economy of forest dwellers. Forest plants provide fruits, roots, tubers, leaves and medicinal plants for forest based sustenance patterns. Forests are the homes to innumerable species of plant and animals and wildlife. They bring about balance in the forest ecosystem and maintain equilibrium between various forms of life. They maintain wide variety of biodiversity.

3. Regulation of Hydrological Cycle: Forested watersheds act like giant sponges since they absorb rain water and slow down water run off. The absorbed water is slowly released for recharge of springs. About 50 to 80 per cent of the moisture in the air, in tropical forests, comes from the transpiration of plants and trees and this moisture brings about rains. Forest plants also help in attracting clouds and causing rains.

4. Moderation of Climate: Forests are thermostat of nature and great moderators of climate. By releasing water vapours from their surfaces, plants increase humidity in the atmosphere. Increased rate of release of water as vapours, brings the atmospheric temperature down. By promoting seepage, forests increase the rate of recharge of ground water. They play a significant role in weather generation and reducing wind velocity. Forests also cause rainfall. Trees along rivers, streams, lakes and other water bodies lower water temperature and temperature of the atmosphere. Large trees provide shade to trespassers.

Among tangible benefits, the productive value of Indian forests is not very encouraging but their regulatory and protective values can not be ignored.

Aesthetic Value of Forests

Forests have a lot of aesthetic value. People all over the world appreciate beauty and tranquillity of forests. Forests are the centres of recreation, joy and spiritual renewal. They provide pleasant weather, pure and fresh air and beautiful scenery and thus refresh both body and mind. Every year lakhs of people visit national parks, sanctuaries and innumerable picnic spots in the forest areas all over the world for recreation. The joyful thrill of spotting a wild animal in the forest can be rightly appreciated only by those who have experienced it. Hill stations which attract tourists not only from India but also from abroad owe most of their charm to the existence of dense forests. India being very rich in its forests and wildlife has a great tourist potential and has a vast scope of eco-tourism.

DEFORESTATION

Deforestation refers to removal of trees or other such plants from a land area without intention of reforesting it. It started long back when man invented fire. Man also exploited forests for food, shelter, agriculture and to meet his other needs. Considerable forests were felled during First and the Second World War to meet the needs of war. In India, tribals in North-East Hill States and some part of Central India started *jhum cultivation* or *shifting cultivation, i.e.*, slash and burn the forests for agricultural land, discard it after two or three years and then move to

Causes of Deforestation

Causes of deforestation are many. Some outstanding causes of deforestation are:

1. Population Pressure for Agricultural Land: Unprecedented growth in the world population is demanding increased agricultural production to

sustain high birth ratio. In developing tropical countries there is a trend of conversion of forests to agricultural land and land for other uses. Shifting cultivation (*jhum*), a 9000-year old practice, is a more specific cause of deforestation. This land use changes pattern of slash and burn for agriculture, practised by landless indigenous people in the tropics, is causing a great damage to the forests. The percentage of this practice is about 50 in South-East Asia including India where it is common in northern hill states and some parts of Madhya Pradesh. Local tribal people practise this type of agriculture in which they clear some forest area, cultivate it for two or three years, abandon it and move to new areas of the forest. Abandoned forest patches (*forest fallows*) may revert back to forests, if left undistributed. But, due to population pressure, little of forest fallow is allowed to revert back. It is estimated that about 50 million people that constitute about 10 per cent of world's population and about 240 million hectares of closed forests are involved in this practice and the rate is increasing at an annual average of 1.25 per cent each year.

2. Fuelwood Requirement: Fuelwood demand is tremendously increasing in the poor people of developing tropical countries. As such, several million hectares of forest land has to be stripped to meet this demand. About 1.5 billion people world over largely depend on fuelwood for their energy requirement. In India, demand for fuelwood has increased sharply. Rapid population growth has caused the demand of fuelwood to shoot up from 300 to 500 million tons in 2001. In India, maximum use of wood is as fuelwood.

3. Industrial Demand for Wood: Forests provide wood for construction of houses, making furniture, plywood, packing cases, pulp for paper industry, match box, pencils *etc*. At present plywood is in great demand for making doors, windows, packing cases for tea and apple, furniture *etc*. In India, industrial requirement of wood is estimated to be about 40 million m^3, out of which about 50 per cent is met by bamboo plantation. It has been observed that poorly performed commercial logging causes degradation of the forests.

4. Development Projects: Forests have undergone massive destruction for various development projects, such as dams, river valley projects, reservoirs, hydroelectric projects, construction of highways, road, rail roads, irrigation canals *etc*. Dams, hydroelectric projects and reservoirs submerge large forest tracts causing harm to the plants and animals of the area. They uproot thousands of forest dwellers not only from their area of residence, but also from their livelihood. Much before construction of these development projects, a lot of forest land is cleared for residence of their workers for which wood and other produce are used up.

5. Overgrazing: The poor in the tropics mainly rely on clearing the forest land to convert it into grazing land for their cattle. Overgrazing leads to further degradation of these lands. India has a large livestock population but grazing land is limited to only 13 million hectares. This much of grazing land is insufficient to meet the demand and as such the livestock naturally grazes in the forests and causes a lot of grazing pressure and loss of the forests. The animals trample the seedlings and cause compaction of soil so that its water storing capacity is decreased and run off capacity is increased. The combined effect of these activities leads to soil erosion and deforestation.

6. Quarrying and Mining Operations: In forested areas, especially in hilly regions, quarrying and mining operations are extremely harmful as they cause great damage to vegetation over large areas due to mine dusts, transportation of ore and mine wastes. In India, large forest land areas are clear felled and laid barren as a result of open cast mining of mica, coal, manganese, limestone *etc*.

7. Forest fires: Sometimes forest suffers from forest fires which may be natural or man-made. A forest fire is any unexpected or unregulated combustion of vegetation spreading over a forest area. The forest fires are of three types:

(*i*) **Ground Fire:** The fire spreads when dry grasses, lichens or top dry humus *etc*., burn up. Usually ground fire does not damage full grown trees, but poses the danger of transforming it into a crown fire. It normally destroys herbs, shrubs and bushes. Ground fire initiates usually after a prolonged drought of the forest floor during second half of the summer.

(*ii*) **Tree Crown Fire:** The fire in the upper portion of the tree is normally derived from ground fire as there appears no reason of combustion of tree canopies without the support of ground fire.

(*iii*) **Underground Fire:** The fire results due to ground fire. Peat soils, which have undergone prolonged drought in the forests are prone to ground fire. After the upper peat soil cover is burnt, smouldering continues in patches covering dry earth mounds and the fire penetrates into the peat horizon of the soil. Underground fire results funnel-shaped pits. Since this fire spreads under the ground in the horizontal manner, tree roots are damaged.

Most of the forests fires are deliberate burning of trees by people in timber trade with the conspiracy of forest guards so that the burnt trees are auctioned at cheaper rate and traders may buy burnt trees.

Forest fires usually start by lightning and release the minerals locked in the organic matter. The mineral rich ashes, necessary for the growth of